CleanTech Global Innovations & Grid Transition 2024

Section 1: Next-Generation Perovskite Cells

Perovskite tandem cells have achieved laboratory conversion efficiencies exceeding 33.9% in 2024. Commercial durability tests demonstrate over 25 years of operational stability.

Section 2: High-Voltage Direct Current (HVDC) Supergrids

Long-distance HVDC transmission lines now allow ultra-low transmission losses of under 3% per 1,000 kilometers, enabling intercontinental clean energy sharing.